

Advance Java Solution - 3

1. Write a Java program to insert data on database using JDBC.

```
// add_data.java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class add_data extends JFrame implements ActionListener{

    JTextField txt_id, nm, mob, ct, addrs;
    JButton btn;

    //constructor
    public add_data(){

        // for Frame
        setTitle("MS Access Insert Form");
        setSize(400, 350);
        setLayout(new GridLayout(6, 2, 10, 10));

        // Labels and TextFields
        add(new JLabel("ID:"));
        txt_id = new JTextField();
        add(txt_id);

        add(new JLabel("Name:"));
        nm = new JTextField();
        add(nm);

        add(new JLabel("Mobile:"));
        mob = new JTextField();
        add(mob);

        add(new JLabel("City:"));
        ct = new JTextField();
        add(ct);

        // Label and TextFields
        add(new JLabel("Address:"));
        addrs = new JTextField();
        add(addrs);

        // Button
        btn = new JButton("Insert");
        add(new JLabel());
        add(btn);

        btn.addActionListener(this); // Event Call

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // close window
        setLocationRelativeTo(null); // center screen
        setVisible(true);
    } // end of constructor

    // Function code
    public void actionPerformed(ActionEvent e) {
        try{
            // Driver
            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
            // Connection
            Connection con =
DriverManager.getConnection("jdbc:ucanaccess://E:/JavaProject/student_db.accdb");

            // Query
            String sql = "INSERT INTO Table1 (id, name, mobile, city, address) VALUES (?, ?, ?, ?, ?)";
```

```

        PreparedStatement ps = con.prepareStatement(sql);
        ps.setInt(1, Integer.parseInt(txt_id.getText()));
        ps.setString(2, nm.getText());
        ps.setString(3, mob.getText());
        ps.setString(4, ct.getText());
        ps.setString(5, addrs.getText());

        int rows = ps.executeUpdate(); //Query Run
        if (rows > 0){
            JOptionPane.showMessageDialog(this, "Record Inserted Successfully!");
            // Clear fields after insert
            txt_id.setText("");
            nm.setText("");
            mob.setText("");
            ct.setText("");
            addrs.setText("");
        }
        con.close();
    }
    // Error Handling
    catch (Exception ex){
        JOptionPane.showMessageDialog(this, "Error: " + ex.getMessage());
    }
}
public static void main(String[] args) {
    new add_data(); // constructor called
}
}

```

2. Write a Java pogram to display data from database using JDBC with above database, table.

```

// Display Data using JTable
// display.java
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.sql.*;

public class display extends JFrame {
    JTable tbl; // display data on JTable
    DefaultTableModel model;
    //constructor
    public display(){

        setTitle("Display Data");
        setSize(700, 400);
        setLocationRelativeTo(null); // Screen on center
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        model = new DefaultTableModel();
        tbl = new JTable(model);
        // Column Names / Title
        model.addColumn("sid");
        model.addColumn("name");
        model.addColumn("gender");
        model.addColumn("course");
        model.addColumn("mobile no");

        add(new JScrollPane(tbl), BorderLayout.CENTER);
        loadData(); // UDF Call
        setVisible(true);
    }
    // UDF Code
    public void loadData(){
        try {
            // Load Driver
            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
            // Connection
            Connection con = DriverManager.getConnection
            ("jdbc:ucanaccess://E:/JavaProject/student_db.accdb");

```

```

        String sql = "SELECT * FROM student_tbl";
        PreparedStatement ps = con.prepareStatement(sql);
        ResultSet rs = ps.executeQuery(); // Query Execution

        while (rs.next()) {
            model.addRow(new Object[]{
                rs.getInt("sid"),
                rs.getString("name"),
                rs.getString("gender"),
                rs.getString("course"),
                rs.getString("mobile_no")
            });
        }
        con.close();
    }

    catch (Exception e) {
        JOptionPane.showMessageDialog
            (this, "Error: " + e.getMessage());
    }
} // end of constructor
public static void main(String[] args){
    new display(); // Constructor Called
}
} //end

```

3. Write a Java program to delete the data based on sid from above database, table.

```

// Delete Data
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class delete_data2 extends JFrame implements ActionListener {

    JLabel lblId;
    JTextField txt_id;
    JButton btn;
    Connection con;
    // Constructor
    delete_data2() {
        // For Frame
        setTitle("Delete Data Form");
        setSize(400, 350);
        setLayout(new GridLayout(6, 2, 10, 10));

        //Label and Textfield
        add(new JLabel("Enter Student ID:"));
        txt_id = new JTextField();
        add(txt_id);

        // Button
        btn = new JButton("Delete Data");
        add(new JLabel());
        add(btn);

        btn.addActionListener(this); // Function Call

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }
    // Button Event,Function Code
    public void actionPerformed(ActionEvent e) {
        try {
            // Driver
            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
            // Connection

```

```

        Connection con =
DriverManager.getConnection("jdbc:ucanaccess://E:/JavaProject/student_db.accdb");

        int id = Integer.parseInt(txt_id.getText());

        // Query
        String sql = "DELETE FROM Table1 WHERE id=?";
        PreparedStatement pst = con.prepareStatement(sql);

        pst.setInt(1, id);

        int rows = pst.executeUpdate();

        if(rows > 0){
            JOptionPane.showMessageDialog(this,"Data Deleted Successfully!");
            txt_id.setText("");
        }
        else{
            JOptionPane.showMessageDialog(this,"Data Not Found!");
        }
    }
    // Error Handling
    catch(Exception ex) {
        JOptionPane.showMessageDialog(this,ex);
    }
} // End of Constructor

public static void main(String[] args) {
    new delete_data2(); // Constructor Call
}
} // end

```

4. Write a Java program to update data based on sid, update student name and course from above database and table.

```

// Update Data
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class update_data2 extends JFrame implements ActionListener {

    JLabel lblId, lblName, lblCity;
    JTextField txtId, txtName, txtCity;
    JButton btn_update;
    Connection con;

    // Constructor
    update_data2() {

        // for Frame
        setTitle("Update Data Form");
        setSize(400, 350);
        setLayout(new GridLayout(6, 2, 10, 10));

        // Labels
        lblId = new JLabel("Enter Student ID:");
        add(lblId);

        lblName = new JLabel("Name:");
        add(lblName);

        lblCity = new JLabel("City:");
        add(lblCity);

        // TextFields
        txtId = new JTextField();
        add(txtId);
    }
}

```

```

        txtName = new JTextField();
        add(txtName);

        txtCity = new JTextField();
        add(txtCity);

        // Button
        btn_update = new JButton("Update Record");

        add(btn_update);

        btn_update.addActionListener(this); // function call

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    // Update Button Event // function code
    public void actionPerformed(ActionEvent e) {
        try {
            // Driver
            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
            // Connection
            Connection con =
DriverManager.getConnection("jdbc:ucanaccess://E:/JavaProject/student_db.accdb");

            int id = Integer.parseInt(txtId.getText());
            String name = txtName.getText();
            String city = txtCity.getText();
            //Query
            String sql = "UPDATE Table1 SET name=?, city=? WHERE id=?";
            PreparedStatement pst = con.prepareStatement(sql);

            pst.setString(1, name);
            pst.setString(2, city);
            pst.setInt(3, id);

            int rows = pst.executeUpdate();

            if(rows > 0)
                JOptionPane.showMessageDialog(this,"Data Updated Successfully!");
            else
                JOptionPane.showMessageDialog(this,"Data Not Found!");

        }
        // Error Handling
        catch(Exception ex) {
            JOptionPane.showMessageDialog(this,ex);
        }
    } // End of Constructor

    public static void main(String[] args) {
        new update_data2(); // Constructor call
    }
} //end

```

5. Write a Java program to perform Insert and Display operation for following fields.

➔ insert_a3p5.javac

```

// insert_a3p5.java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class insert_a3p5 extends JFrame implements ActionListener{

```

```

JTextField iid, product_name, price, qty, discount_rs, SGST_rs, CGST_rs, total_amount;
JButton btn;

//construcourseor
public insert_a3p5(){

    // for Frame
    setTitle("MS Access Insert Form");
    setSize(400, 350);
    setLayout(new GridLayout(6, 2, 10, 10));

    // Labels and TextFields
    add(new JLabel("iid : "));
    iid = new JTextField();
    add(iid);

    add(new JLabel("product_name : "));
    product_name = new JTextField();
    add(product_name);

    add(new JLabel("price : "));
    price = new JTextField();
    add(price);

    add(new JLabel("qty : "));
    qty = new JTextField();
    add(qty);

    add(new JLabel("discount_rs :"));
    discount_rs = new JTextField();
    add(discount_rs);

    add(new JLabel("SGST_rs :"));
    SGST_rs = new JTextField();
    add(SGST_rs);

    add(new JLabel("CGST_rs :"));
    CGST_rs = new JTextField();
    add(CGST_rs);

    add(new JLabel("total_amount :"));
    total_amount = new JTextField();
    add(total_amount);

    // Button
    btn = new JButton("Insert");
    add(new JLabel());
    add(btn);

    btn.addActionListener(this); // Event Call

    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // close window
    setLocationRelativeTo(null); // center screen
    setVisible(true);
} // end of construcourseor

// Funcourseion code
public void actionPerformed(ActionEvent e) {
    try{
        // Driver
        Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
        // Connecourseion
        Connection con =
DriverManager.getConnection("jdbc:ucanaccess://E:/JavaProjecourse/invoice_db.accdb");

        // Query
        String sql = "INSERT INTO invoice_tbl (iid, product_name, price, qty, discount_rs, SGST_rs,
CGST_rs, total_amount) VALUES (?, ?, ?, ?, ?, ?, ?, ?)";

        PreparedStatement ps = con.prepareStatement(sql);
        ps.setInt(1, Integer.parseInt(iid.getText()));
        ps.setString(2, product_name.getText());

```

```

        ps.setString(3, price.getText());
        ps.setString(4, qty.getText());
        ps.setString(5, discount_rs.getText());
        ps.setString(6, SGST_rs.getText());
        ps.setString(7, CGST_rs.getText());
        ps.setString(8, total_amount.getText());

        int rows = ps.executeUpdate(); //Query Run
        if (rows > 0){
            JOptionPane.showMessageDialog(this, "Record Insert Successfully!");
            // Clear fields after insert
            iid.setText("");
            product_name.setText("");
            price.setText("");
            qty.setText("");
            discount_rs.setText("");
            SGST_rs.setText("");
            CGST_rs.setText("");
            total_amount.setText("");
        }
        con.close();
    }
    // Error Handling
    catch (Exception ex){
        JOptionPane.showMessageDialog(this, "Error: " + ex.getMessage());
    }
}
public static void main(String[] args) {
    new insert_a3p5(); // construcourseor called
}
}

```

➔ display_a3p5.java

```

// display_a3p5.java
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.sql.*;
public class display_a3p5 extends JFrame {
    JTable tbl; // display_a3p5 data on JTable
    DefaultTableModel model;
    //constructor
    public display_a3p5(){

        setTitle("Display Data");
        setSize(700, 400);
        setLocationRelativeTo(null); // Screen on center
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        model = new DefaultTableModel();
        tbl = new JTable(model);
        // Column Names / Title
        model.addColumn("iid");
        model.addColumn("product_name");
        model.addColumn("price");
        model.addColumn("qty");
        model.addColumn("discount_rs");
        model.addColumn("SGST_rs");
        model.addColumn("CGST_rs");
        model.addColumn("total_amount");

        add(new JScrollPane(tbl), BorderLayout.CENTER);
        loadData(); // UDF Call
        setVisible(true);
    }
    // UDF Code
    public void loadData(){
        try {
            // Load Driver

```

```

        Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
        // Connection
        Connection con = DriverManager.getConnection
        ("jdbc:ucanaccess://E:/JavaProject/invoice_db.accdb");

        String sql = "SELECT * FROM invoice_tbl";
        PreparedStatement ps = con.prepareStatement(sql);
        ResultSet rs = ps.executeQuery(); // Query Execution

        while (rs.next()) {
            model.addRow(new Object[]{
                rs.getInt("iid"),
                rs.getString("product_name"),
                rs.getString("price"),
                rs.getString("qty"),
                rs.getString("discount_rs"),
                rs.getString("SGST_rs"),
                rs.getString("CGST_rs"),
                rs.getString("total_amount")
            });
        }
        con.close();
    }

    catch (Exception e) {
        JOptionPane.showMessageDialog
        (this, "Error: " + e.getMessage());
    }
} // end of constructor
public static void main(String[] args){
    new display_a3p5(); // Constructor Called
}
} //end

```

6. Write a Java program to perform Delete and Update operation for following fields.

➔ delete_a3p6.java

```

// delete_a3p6 Data
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class delete_a3p6 extends JFrame implements ActionListener {

    JLabel lblId;
    JTextField sid;
    JButton btn;
    Connection con;
    // Constructor
    delete_a3p6() {
        // For Frame
        setTitle("Delete Data Form");
        setSize(400, 350);
        setLayout(new GridLayout(6, 2, 10, 10));

        //Label and Textfield
        add(new JLabel("Enter employee ID:"));
        sid = new JTextField();
        add(sid);

        // Button
        btn = new JButton("Delete Data");
        add(new JLabel());
        add(btn);

        btn.addActionListener(this); // Function Call
    }
}

```

```

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }
    // Button Event,Function Code
    public void actionPerformed(ActionEvent e) {
        try {
            // Driver
            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
            // Connection
            Connection con =
DriverManager.getConnection("jdbc:ucanaccess://E:/JavaProject/employee_db.accdb");

            int id = Integer.parseInt(sid.getText());

            // Query
            String sql = "Delete FROM emp_tbl WHERE id=?";
            PreparedStatement pst = con.prepareStatement(sql);

            pst.setInt(1, id);

            int rows = pst.executeUpdate();

            if(rows > 0){
                JOptionPane.showMessageDialog(this,"Data Deleted Successfully!");
                sid.setText("");
            }
            else{
                JOptionPane.showMessageDialog(this,"Data Not Found!");
            }
        }
        // Error Handling
        catch(Exception ex) {
            JOptionPane.showMessageDialog(this,ex);
        }
    } // End of Constructor

    public static void main(String[] args) {
        new delete_a3p6(); // Constructor Call
    }
} // end

```

➔update_a3p6.java

```

// Update Data
//update_a3p6.java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class update_a3p6 extends JFrame implements ActionListener {

    JLabel lblleid, lblname, lbldesignation, lblbasic_salary, lblta, lblda, lblhra, lblpf, lblnet_salary;
    JTextField txteid, txtname, txtdesignation, txtbasic_salary, txtta, txtda, txthra, txtpf,
txtnet_salary;
    JButton btn_update;
    Connection con;

    // Constructor
    update_a3p6() {

        // for Frame
        setTitle("Update Data Form");
        setSize(400, 350);
        setLayout(new GridLayout(6, 2, 10, 10));

        // Labels
        lblleid = new JLabel("Enter Employee ID : ");
    }
}

```

```

        add(lblleid);

        lblname = new JLabel("Enter Name : ");
        add(lblname);

        lbldesignation = new JLabel("Enter Designation : ");
        add(lbldesignation);

        lblbasic_salary = new JLabel("Enter Basic Salary : ");
        add(lblbasic_salary);

        lblta = new JLabel("Enter TA : ");
        add(lblta);

        lblda = new JLabel("Enter DA : ");
        add(lblda);

        lblhra = new JLabel("Enter HRA : ");
        add(lblhra);

        lblpf = new JLabel("Enter PF :");
        add(lblpf);

        lblnet_salary = new JLabel("Enter Net Salary : ");
        add(lblnet_salary);

        // TextFields
        txteid = new JTextField();
        add(txteid);

        txtname = new JTextField();
        add(txtname);

        txtdesignation = new JTextField();
        add(txtdesignation);

        txtbasic_salary = new JTextField();
        add(txtbasic_salary);

        txtta = new JTextField();
        add(txtta);

        txtda = new JTextField();
        add(txtda);

        txthra = new JTextField();
        add(txthra);

        txtpf = new JTextField();
        add(txtpf);

        txtnet_salary = new JTextField();
        add(txtnet_salary);

        // Button
        btn_update = new JButton("Update Record");

        add(btn_update);

        btn_update.addActionListener(this); // function call

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    // Update Button Event // function code
    public void actionPerformed(ActionEvent e) {
        try {
            // Driver
            Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");
            // Connection

```

```

        Connection con =
DriverManager.getConnection("jdbc:ucanaccess://E:/JavaProject/employee_db.accdb");

        int eid = Integer.parseInt(txteid.getText());
        String name = txtname.getText();
        String designation = txtdesignation.getText();
        int basic_salary = Integer.parseInt(txtbasic_salary.getText());
        int ta = Integer.parseInt(txtta.getText());
        int da = Integer.parseInt(txtda.getText());
        int hra = Integer.parseInt(txthra.getText());
        int pf = Integer.parseInt(txtpf.getText());
        int net_salary = Integer.parseInt(txtnet_salary.getText());

        //Query
        String sql = "UPDATE emp_tbl SET name=?, designation=?, basic_salary=?, ta=?, da=?, hra=?, pf=?,
net_salary=? WHERE eid=?";
        PreparedStatement pst = con.prepareStatement(sql);

        pst.setString(1, name);
        pst.setString(2, designation);
        pst.setInt(3, basic_salary);
        pst.setInt(4, ta);
        pst.setInt(5, da);
        pst.setInt(6, hra);
        pst.setInt(7, pf);
        pst.setInt(8, net_salary);

        int rows = pst.executeUpdate();

        if(rows > 0)
            JOptionPane.showMessageDialog(this,"Data Updated Successfully!");
        else
            JOptionPane.showMessageDialog(this,"Data Not Found!");

    }
    // Error Handling
    catch(Exception ex) {
        JOptionPane.showMessageDialog(this,ex);
    }
} // End of Constructor

public static void main(String[] args) {
    new update_a3p6(); // Constructor call
}
} //end

```